Additional file 3: Supplementary Sensitivity Tables

Additional file 3. Supplementary Sensitivity Tables

Supplementary tables for the manuscript *"A Multiple Correspondence Analysis Approach to Mitigate the Non-Communicable Disease (NCD) Epidemic: Introducing a Health-Related Behavior-Based NCD Vulnerability Index."*

Table S1. Multilevel estimator comparison on the ten-input primary NCDVI

The primary specification (`glmer` with the sample weight as a frequency weight) is the literature-matched estimator. The supplementary specification (`WeMix::mix` pseudo maximum likelihood with cluster-scaled level-one weight) is the fully survey-design-corrected estimator.

Specification	Tier 5 OR	95% CI	Var(u)	ICC
glmer, weights as frequency weights (primary)	15.79	[13.12, 19.00]	0.137	0.040
WeMix::mix, scaled level-one weight (supplementary)	27.10	[21.54, 34.08]	1.609	0.329

Table S2. MCA design alternatives

Each row varies a single design choice. The production specification is the first row.

Variant	Tier 5 OR	CV AUC	Monotonic?
ncp=1, quintile cut, 5 tiers, MCA (production)	15.79	0.793	Yes
ncp=2	14.6	0.791	Yes
ncp=3	12.4	0.789	No
Equal-width cut	18.2	0.793	Yes
Jenks natural-break cut	16.5	0.792	Yes
k-means cut	17.1	0.793	Yes
4 tiers	12.0	0.793	Yes
6 tiers	18.0	0.793	Yes
7 tiers	20.1	0.793	Yes
PCAmix instead of MCA	15.8	0.793	Yes
Cooking location removed	15.79	0.793	Yes

Table S3. Endogeneity sensitivity: Tier 5 odds ratio across NCDVI versions

NCDVI	Inputs	Tier 5 OR	95% CI	Tier 5 collapse vs ten-input
Ten-input primary	All 10	15.79	[13.12, 19.00]	reference
Six-input sensitivity, 2017-18	Structural only	3.22	[2.64, 3.92]	-80%
Six-input sensitivity, BDHS 2022, 5-tier	Structural only	4.03	[1.78, 9.11]	-75%
Six-input sensitivity, BDHS 2022, 4-tier collapsed	Structural only	3.73	[1.90, 7.32]	-76%

Table S4. Discrimination ceiling: M0 to M6 build-up and gradient-boosted comparison

Model	Cross-validated AUC
M0: confounders only	0.740
M1: M0 plus categorical NCDVI tier	0.793
M2: M1 with continuous NCDVI score	0.806
M3: M2 plus continuous BMI	0.812
M4: M3 plus age-by-sex interaction	0.814
M5: M4 plus sex-by-BMI interaction	0.815
M6: M5 plus natural splines on age and BMI	0.816
Gradient-boosted tree on the M6 feature set	0.811